

ALEXANDER CITY, AL, USA

WMO: 720265

Lat: 32.915N

Lon: 85.963W

Elev: 686

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63833

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.5	27.9	4.9	7.3	39.8	10.5	9.6	35.7	19.4	44.9	17.7	45.0	4.0	350	0.343

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	95.2	75.5	93.0	75.8	91.1	75.4	81.0	88.8	79.7	87.6	78.2	86.4	5.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	154.3	85.2	77.4	146.3	83.5	76.5	141.6	82.4	45.1	89.0	43.8	87.5	42.2	86.6	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	12.9	11.3	16.7	95.7	4.1	2.4	13.7	97.5	11.3	98.9	9.0	100.2	6.0	102.0
			15.1	82.2	4.0	2.1	12.2	83.7	9.9	84.9	7.6	86.0	4.7	87.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	45.7	50.1	56.4	63.5	71.2	78.8	80.8	80.2	76.1	65.0	54.2	49.2
	DBStd	14.32	9.87	8.97	8.43	6.56	5.87	3.77	3.00	3.37	5.24	7.91	7.83	8.95
	HDD50	524	204	103	38	2	0	0	0	0	0	3	44	130
	HDD65	2359	599	423	283	104	18	0	0	0	3	99	335	495
	CDD50	5757	71	105	236	407	656	863	956	937	783	470	169	104
	CDD65	2117	1	5	17	59	209	413	491	472	336	100	10	4
	CDH74	19119	5	15	171	597	1856	3918	4687	4285	2757	753	61	14
	CDH80	7875	0	0	19	91	635	1797	2129	1904	1116	179	5	0
Wind	WSAvg	3.6	4.6	4.7	4.9	4.3	3.3	2.7	2.6	2.6	2.8	3.3	3.7	4.1
Precipitation	PrecAvg	51.4	5.2	5.0	5.7	4.4	3.6	4.5	5.3	3.9	3.7	2.5	4.4	4.3
	PrecMax	62.2	9.1	8.8	12.6	8.1	9.8	9.6	10.3	6.6	7.9	9.0	11.1	7.0
	PrecMin	29.7	2.4	2.2	1.3	2.3	0.2	1.9	1.7	2.0	0.8	0.2	2.2	2.8
	PrecStd	7.5	1.9	1.9	2.8	1.6	2.2	2.4	2.2	1.5	1.9	2.0	2.2	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.7	76.6	82.3	87.0	91.4	97.4	97.0	96.1	95.3	89.8	79.7	76.0
		MCWB	63.2	66.2	61.7	66.7	71.9	74.0	76.7	75.4	75.7	73.8	67.4	68.3
	2%	DB	69.6	72.7	78.9	82.3	89.7	94.3	94.5	94.8	91.7	84.3	75.2	71.5
		MCWB	62.3	62.7	62.0	64.7	71.4	74.6	76.4	76.1	74.2	70.6	63.0	64.7
	5%	DB	67.5	69.2	75.1	80.5	87.7	91.5	92.0	91.4	90.2	81.2	72.6	68.3
		MCWB	60.7	60.1	60.6	64.1	71.1	75.2	76.3	76.3	75.3	67.8	63.2	63.8
	10%	DB	62.0	66.3	72.0	77.7	83.1	89.8	90.5	90.1	87.8	78.7	69.4	64.7
		MCWB	57.1	59.2	59.2	63.4	69.0	74.4	76.3	76.7	73.8	66.6	60.6	60.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.2	68.8	68.3	72.7	76.4	81.4	82.2	83.2	80.5	76.1	72.6	70.4
		MCDB	70.0	73.8	72.2	80.0	84.1	89.0	89.8	90.6	88.7	82.7	74.6	73.5
	2%	WB	64.6	66.3	66.6	70.2	74.6	79.5	80.6	81.1	78.6	74.2	69.0	67.7
		MCDB	68.3	70.0	72.0	76.9	83.6	87.0	88.7	88.8	86.5	80.3	72.5	69.6
	5%	WB	61.7	63.8	64.8	68.2	73.2	77.7	79.1	79.7	77.2	72.5	66.2	65.4
		MCDB	65.0	67.4	71.1	74.9	81.8	86.6	87.3	87.7	84.6	77.6	70.4	68.0
	10%	WB	58.2	60.1	62.5	66.7	71.9	76.4	77.8	78.2	75.5	70.4	62.7	60.8
		MCDB	61.8	63.7	70.3	73.6	79.8	85.6	86.4	85.9	82.4	75.1	66.9	64.2
Mean Daily Temperature Range	5% DB	MDBR	20.9	21.4	22.9	24.0	21.7	19.8	18.4	17.7	19.2	22.3	23.8	19.6
		MCDBR	23.3	24.5	26.4	26.1	25.9	22.5	20.7	20.2	23.0	24.6	25.9	21.1
		MCWBR	16.5	15.5	13.1	12.3	11.0	9.3	8.1	8.5	9.2	12.0	15.8	16.0
	5% WB	MCDBR	18.5	19.0	19.9	20.0	20.0	19.5	18.4	17.8	18.6	17.8	18.8	17.6
	MCWBR	15.5	15.1	13.1	12.0	9.9	9.7	8.9	8.9	8.5	10.3	15.2	15.8	
Clear-Sky Solar Irradiance	taub	0.318	0.323	0.354	0.386	0.446	0.477	0.524	0.521	0.423	0.368	0.341	0.322	
	taud	2.527	2.512	2.439	2.337	2.211	2.175	2.054	2.052	2.317	2.458	2.472	2.529	
	Ebn at Noon	286	296	293	286	269	259	246	245	267	276	274	278	
	Edh at Noon	27	30	35	40	46	47	53	53	39	31	28	25	
All-Sky Solar Radiation	RadAvg	867	1042	1402	1772	1945	1948	1895	1746	1534	1299	976	740	
	RadStd	58	129	125	122	152	149	130	100	136	166	96	80	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+187

Nomenclature: See separate page